Process-bigraph co-simulation vs the embedded vcell-fvsolver hybrid solver

complete · viva-pde-particle--outcomes · 6 studies · 2026-10-03

Executive summary

Question. Can a process-bigraph co-simulation of a PDE process and a Smoldyn process reproduce the hybrid PDE/particle solver embedded in vcell-fvsolver to within its stochastic error, and at what computational cost?

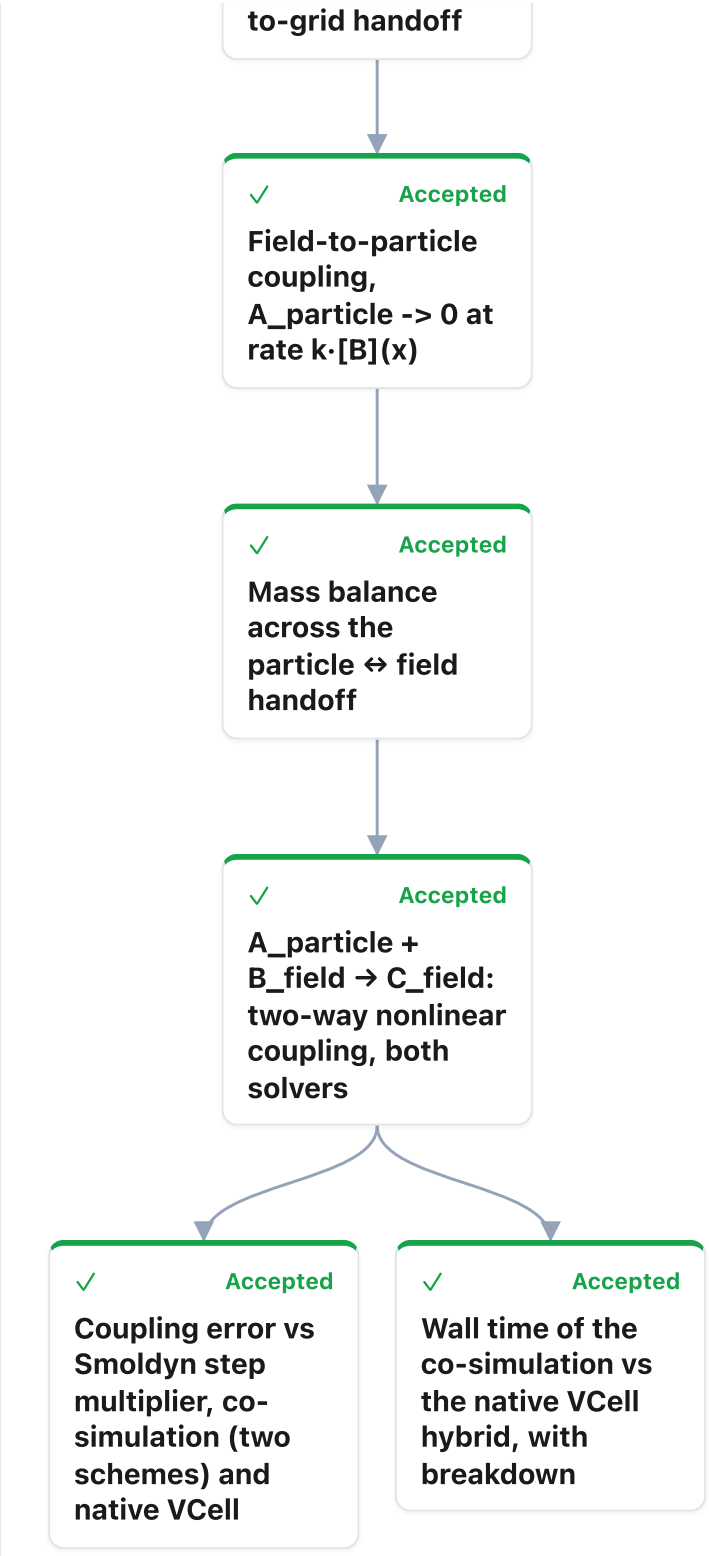
Yes. The process-bigraph co-simulation reproduces the embedded vcell-fvsolver hybrid to within its stochastic error on every like-for-like test, with the same first-order coupling error. - A2: field-to-particle coupling, |z| < 2.6. - A4: two-way bimolecular coupling, |z| < 2.1. - A5: coupling error 1.05× the native error at k = 16. It is also faster: 0.25–0.93× of the native solve time (A6). Where they differ, the native solver is wrong: - A3: boundary cells overproduce field-dependent particle creation by Σfull/Σactual (virtualcell/vcell-fvsolver **24).** - VCell math generation keeps no-op particle processes (virtualcell/vcell **2158). - A second hybrid solve in one process segfaults (vcell-fvsolver#23). - The native solver crashes at a large step multiplier. The co-simulation's "start-of-interval" scheduling halves the coupling error at large k. vcell-fvsolver's hybrid loop fixes a single splitting order.**

Across **6** studies — **6** supported · **0** qualified · **0** refuted · **0** planned. *Supported* = a computational gate passed under the stated model assumptions (evidence of feasibility, not proof the transition occurred).

The investigation graph

The campaign as a dependency graph — each study a node badged with its code-computed status, wired to the studies it builds on.





Status: ✓ Accepted ◐ Investigating ○ Planned ⊘ Blocked ✕ Refuted

Studies

Expand all | Collapse all

Every study in the campaign as a collapsible card — the question it tests, its verdict, and what it found at a glance. Click a card to expand the full record: model, runs, figures, tests, findings, and the limitations that bound it.

01 Each side alone, and the particle-to-grid handoff

ACCEPTED

Before coupling anything: does each half of the co-simulation behave correctly on its own, and does the particle-to-grid handoff (nearest-node binning, boundary element volumes, counts to μM) follow vcell-fvsolver's conventions?

Hypothesis & discriminating control

The FV process reproduces continuum diffusion to discretization accuracy. Smoldyn's binned particle variance grows as 2Dt per axis. Binning conserves molecules exactly. Uniformly distributed particles give a uniform binned concentration up to and including the boundary nodes, because boundary elements have 1/2 (edge) or 1/4 (corner) volume as in VCell's CartesianMesh.

MODEL EXPERIMENT Four checks, all on the shared CartesianGrid: (1) FV diffusion of $\cos(\pi x/L)\cos(\pi y/L)$ (21×21, L = 10 μm, D = 1, dt = 0.01, t = 1 s) against $\exp(-2\pi^2 Dt/L^2)$; (2) 20,000 particles diffusing from the centre cell (D = 1, 4 seeds): the variance of the binned positions, minus $dx^2/6$ (the initial cell spread plus Sheppard's binning correction), against 2Dt; (3) the histogram sum equals the molecule count at every record; (4) 100,000 uniform particles after 2 s of diffusion (11×11, 4 seeds): the counts per node class (corner, edge, interior) match their element-volume shares.

OBSERVABLE `a1_fv_cosine_rel_err, a1_particle_variance_rel_err, a1_histogram_count_mismatch, a1_boundary_class_max_abs_z, a1_corner_to_interior_conc_ratio…`

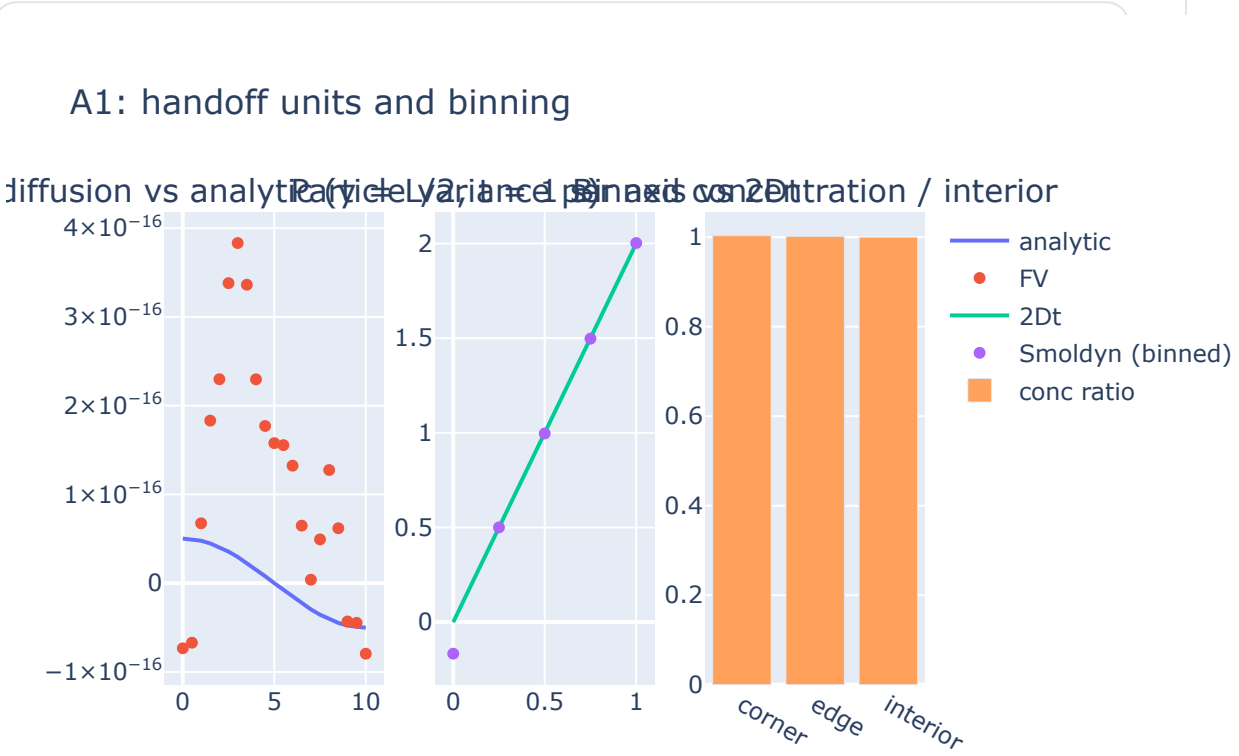
Model & key assumptions

Methods · Simulations

Model	`particle_diffusion`
Sims	1 run
Tests	✓✓✓✓ 4/4 passed

Visualizations

HANDOFF



Finding

F-01 · confirms ● Observed

Each half of the co-simulation is correct in isolation: the FV operator reproduces the analytic cosine-mode decay to 6.0e-4 relative error, and Smoldyn's binned per-axis variance (dx²/6-corrected) matches 2Dt to 0.13% relative error at t=1 s.

from test: fv-matches-continuum-diffusion < `0.01`

runs: `default`

F-02 · confirms ● Observed

The particle-to-grid handoff is exact and volume-correct: the histogram total never differs from the molecule count (mismatch = 0 molecules at every record), and uniformly distributed particles bin to a uniform concentration across node classes (max |z| = 0.48; corner and edge concentrations 1.0034× and 1.0019× interior, both within 0.4%), confirming the 1/4-corner and 1/2-edge element-volume convention.

from test: boundary-volumes-give-uniform-concentration < 4

runs: `default`

Verdict

● PASSED

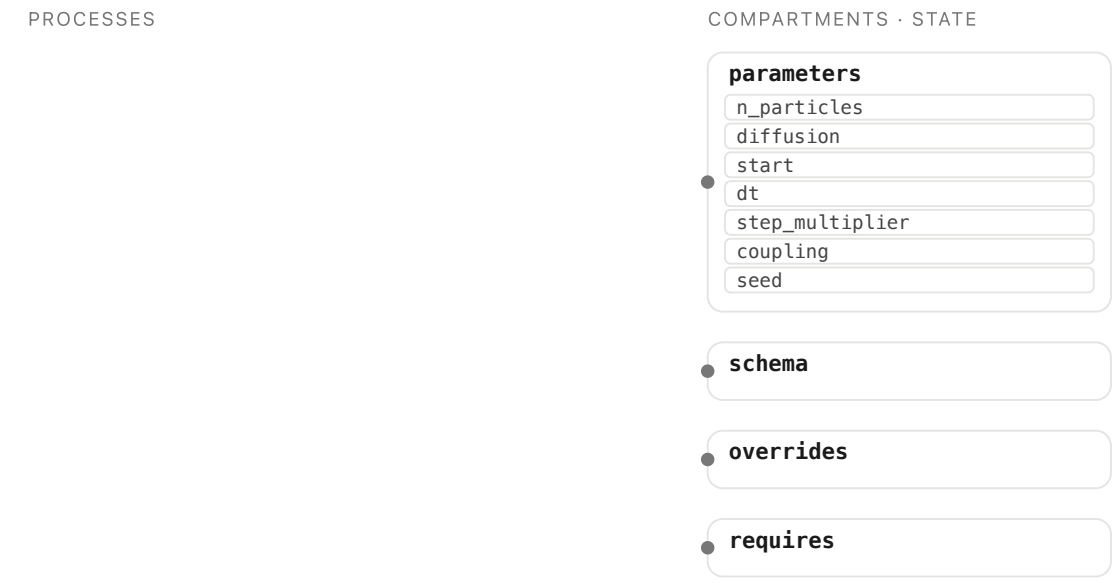
Gate cleared · Accepted

All four checks pass (results/metrics.json, 2026-10-03; re-run after the record-time fix): - FV vs analytic: 6.0e-4 relative error; - binned particle variance vs 2Dt: 0.13% (0.96% before the fix, which was the one-step record lag); - histogram mismatch: 0 molecules; - node

classes vs volume shares: max |z| = 0.48. Corner and edge concentrations are within 0.4% of interior. Both halves and the handoff behave as specified.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.particle_diffusion` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.particle_diffusion`

Simulation run

default

`fv-matches-continuum-diffusion` authored: PASS

`particle-variance-matches-2Dt` authored: PASS

`histogram-conserves-molecules` authored: PASS

`boundary-volumes-give-uniform-concentration` authored: PASS

Reproduce: `vwb run study handoff-units-and-binning`
fv-cosine-mode: `vwb run study handoff-units-and-binning --variant fv-cosine-mode`
particle-variance: `vwb run study handoff-units-and-binning --variant particle-variance`
boundary-volumes: `vwb run study handoff-units-and-binning --variant boundary-volumes`

Tests

Max relative error of the FV cosine mode vs the analytic decay at t = 1 s. `< 0.01` pending

`fv-matches-continuum-diffusion · primary · derived_scalar`

Second order in space and first order in time; expected error ~1e-3 on this grid.

Relative error of the binned per-axis variance (corrected by dx²/6) vs 2Dt at t = 1 s. `particle-variance-matches-2Dt · primary · derived_scalar` Free Brownian motion; 80,000 molecule samples give ~0.5% sampling error on the variance.	`< 0.03`	pending
Max |histogram total - molecule count| over all records. `histogram-conserves-molecules · primary · derived_scalar` Every molecule maps to exactly one node (clamped nearest-node index).	`== 0`	pending
Max |z| over node classes of per-class counts vs element-volume share. `boundary-volumes-give-uniform-concentration · primary · derived_scalar` Binomial sampling; |z| < 4 for a correct volume convention (corner 1/4, edge 1/2).	`< 4`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 02 Field-to-particle coupling, A_particle -> 0 at rate k·[B](x)

● ACCEPTED

Does the Smoldyn process read the PDE field at each molecule as vcell-fvsolver does (nearest node, value from the previous coupling step)? Does the coupled model approach its deterministic continuum limit?

Hypothesis & discriminating control

With a static field and immobile particles, each grid column survives as exp(-k·B·t) exactly in expectation, and nothing decays where B = 0. With diffusing A and B, the ensemble-mean particle concentration matches the all-continuous FV solution of the same model, to within sampling noise.

MODEL EXPERIMENT Part 1: B rises linearly from 0 to 1 µM across x; 4,000 immobile A per run; k = 1/(µM·s); 8 seeds; t = 1 s. Compare per-column survival to exp(-k·B_col·t) with binomial z-scores. Part 2: B is a Gaussian (1 µM peak, D_B = 0.5); 20,000 A with D_A = 1; 8 seeds. Compare the ensemble-mean A (µM) at t = 1 s to continuum_model() solved by FVReactionDiffusion alone (relative L2 over the grid, and total A).

OBSERVABLE `a2_static_field_max_abs_z, a2_zero_field_decays, a2_continuum_rel_l2_err, a2_continuum_total_rel_err,`

a2_native_static_field_max_abs_z…

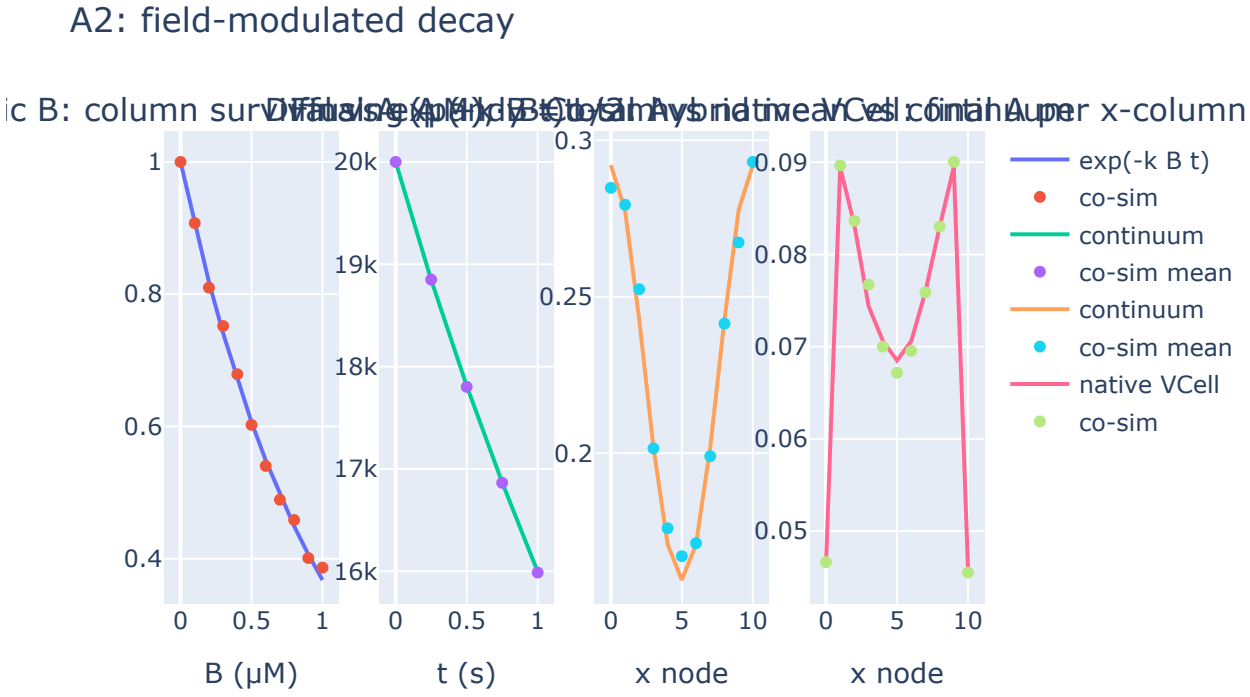
Model & key assumptions

Methods · Simulations

Model	`field_modulated_decay`
Sims	1 run
Tests	✓✓✓✓✓✓✓✓ 8/8 passed

Visualizations

FIELD MODULATED DECAY



Finding

F-01 · confirms · Observed

The particle side reads the PDE field correctly: with static B and immobile A, per-column survival matches N0·exp(-k·B_col·t) with max |z| = 1.56 (co-sim) and 1.68 (native VCell), both well under the |z| < 4 band, and exactly zero molecules decay in the B = 0 column.

from test: survival-follows-local-field < 4

runs: `default`

F-02 · confirms · Observed

The coupled model converges to its deterministic continuum limit: with diffusing A and a Gaussian B, the ensemble-mean A is within 2.7% relative L2 (band 6%) and the total A within 0.05% (band 2%) of the all-continuous FV solution — both at the sampling-noise level.

from test: approaches-continuum-limit < 0.06

runs: `default`

F-03 · confirms · Observed

The co-simulation and VCell's native hybrid are statistically indistinguishable on this field-to-particle coupling: static-field survival agrees at max |z| = 1.37, the diffusing final x-profiles at |z| = 2.54, and total survival to 0.04%, with the B fields identical to 1e-16 — so GridValueProvider's nearest-node field read is reproduced by the process-bigraph decomposition.

from test: cosim-matches-native-diffusing < 4

runs: `default`

Verdict

PASSED

Gate cleared · Accepted

All checks pass (results/metrics.json, 2026-10-03; re-run after the record-time fix). Co-simulation: - column survival vs exp(-k·B·t): max |z| = 1.56; - decays where B = 0: none; - ensemble-mean A vs continuum: 2.7% relative L2, at the sampling-noise level; - total A vs continuum: 0.05%. Co-simulation vs native VCell hybrid (quasi-2D slab 11×11×3, 8 seeds each, matched Poisson initial placement): - native column survival vs exp(-k·B·t): max |z| = 1.68; - co-sim vs native static-field survival: max |z| = 1.37; - B fields identical to 1e-16; - diffusing case, final x-profiles: max |z| = 2.54; - total survival differs by 0.04%. The two implementations are statistically indistinguishable on this model.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals

PROCESSES

COMPARTMENTS · STATE

parameters

- n_particles
- k
- d_a
- d_b
- b_profile
- dt
- step_multiplier
- coupling
- seed

schema

overrides

requires

Static bigraph of `viva_pde_particle.composites.examples.field_modulated_decay` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.field_modulated_decay`

Simulation run

default

`survival-follows-local-field` authored: PASS

`no-decay-where-field-is-zero` authored: PASS

`approaches-continuum-limit` authored: PASS

`total-matches-continuum` authored: PASS

`native-vcell-follows-local-field` authored: PASS

`cosim-matches-native-static` authored: PASS

`cosim-matches-native-diffusing` authored: PASS

`cosim-matches-native-total` authored: PASS

Reproduce: `vwb run study field-modulated-decay`
native-vcell-slab: `vwb run study field-modulated-decay --variant native-vcell-slab`
continuum-limit: `vwb run study field-modulated-decay --variant continuum-limit`

Tests

Max |z| over x-columns of survivors vs N0·exp(-k·B_col·t) (static B, immobile A). — `< 4` pending
`survival-follows-local-field · primary · derived_scalar`
Smoldyn's first-order reaction probability per step is 1-exp(-rate·dt), so survival is exp(-k·B·t) exactly in expectation.

Molecules lost from the B = 0 column. — `== 0` pending
`no-decay-where-field-is-zero · primary · derived_scalar`
Rate k·0 = 0 at the B = 0 nodes.

Relative L2 error of the ensemble-mean A (µM) vs the continuum FV solution at t = 1 s. — `< 0.06` pending
`approaches-continuum-limit · primary · derived_scalar`
Per-node sampling noise of the 8-run mean is about 1/sqrt(8·165), roughly 2.8%; the threshold allows about 2x that.

Relative error of the ensemble-mean total A vs the continuum total at t = 1 s. — `< 0.02` pending
`total-matches-continuum · secondary · derived_scalar`
160,000 total molecule samples give ~0.3% sampling error on the total.

Native VCell hybrid, static B (slab, 8 seeds); max |z| over x-columns of survivors vs N0·exp(-k·B·t). — `< 4` pending
`native-vcell-follows-local-field · primary · derived_scalar`
Same exact expectation as the co-simulation test; validates the headless native reference path.

Two-sample max |z| over x-columns of per-run survival fractions, co-simulation vs native VCell (static B). `cosim_matches_native_static · primary · derived_scalar` Equal expected survival under both solvers; 8 runs each.	< 4	pending
Two-sample max |z| over x-columns of the final A profile, co-simulation vs native VCell (diffusing A and B). `cosim_matches_native_diffusing · primary · derived_scalar` Same model, mesh and coupling scheme in both solvers; differences only from sampling (8 runs each).	< 4	pending
Relative difference of mean total survival, co-simulation vs native VCell (diffusing A and B). `cosim_matches_native_total · secondary · derived_scalar` 160,000 molecule samples per solver; sampling error ~0.5%.	< 0.02	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 03 Mass balance across the particle ↔ field handoff

● ACCEPTED

When molecules move both ways between particle and field representations (A_particle → B_field and B_field → A_particle by field-dependent creation), is total A + B conserved and is the right steady state reached? Do both solvers agree?

Hypothesis & discriminating control

Each side applies only its own half of each mixed reaction, so A + B is conserved in expectation up to the O(dt) lag of the explicit coupling, and total A / total B → k2 / k1. Field-dependent creation must use each grid node's actual cell volume (1/2–1/8 at boundaries); otherwise creation is overproduced by Σ(full cells)/Σ(actual cells).

MODEL EXPERIMENT 11×11×3 slab (10×10×1 µm); k1 = 1 s^{-1}, k2 = 0.5 s^{-1}; B_0 = 0.2 µM (12,044 molecules); D_A = D_B = 1; dt = 0.01; t = 5 s; 16 seeds per solver. Measured: - mass balance and steady-state ratio; - co-sim vs native totals and profiles; - a creation-only experiment (B a static catalyst, 2 s, 8 seeds), comparing created A with the exact expectation k2·[B]·V·602.214·t.

OBSERVABLE `a3_cosim_mass_balance_max_rel_dev, a3_cosim_steady_ratio_rel_err, a3_cosim_creation_ratio, a3_native_creation_ratio, a3_native_creation_vs_bug_factor…`

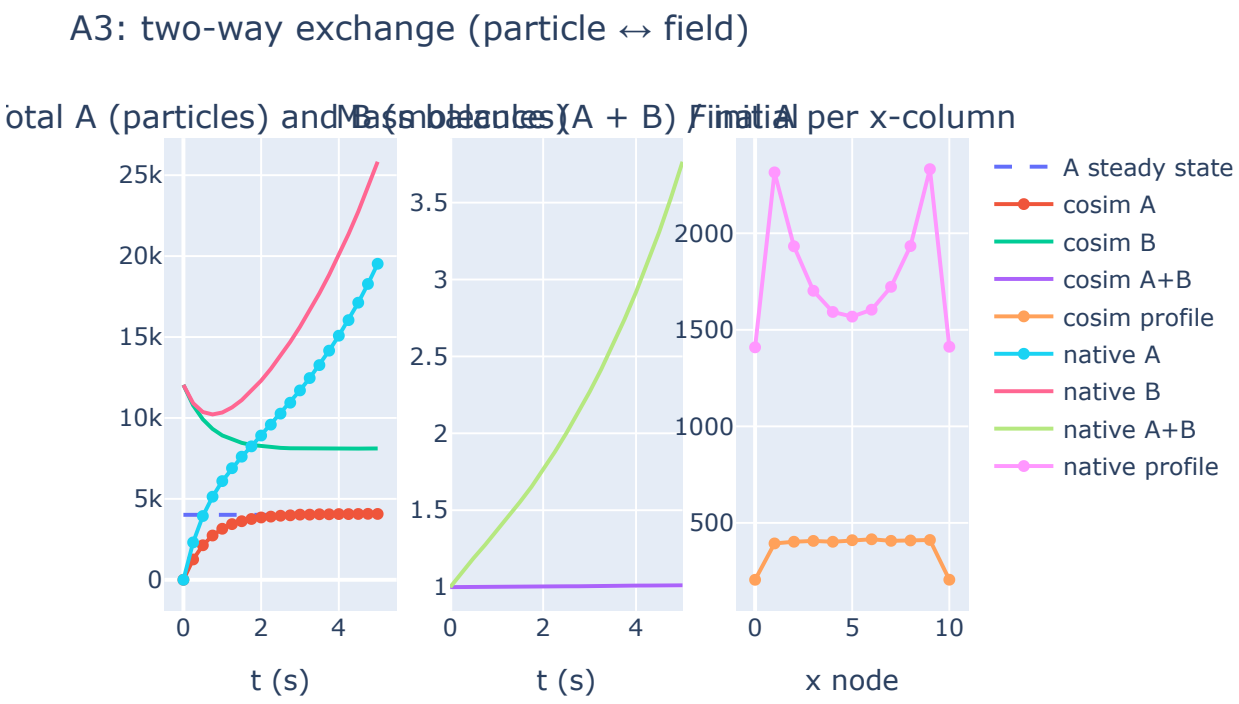
Model & key assumptions

Methods · Simulations

Model	`two_way_exchange`
Sims	1 run
Tests	✓✓✓✓ 4/4 passed

Visualizations

TWO WAY EXCHANGE



Finding

F-01 · confirms · Observed

The co-simulation conserves mass and reaches the right steady state across the two-way handoff: ensemble-mean (A+B) stays within 1.18% of its initial value over 5 s, and the late-time total-A/total-B ratio is within 0.32% of k2/k1 (16 seeds).

from test: cosim-conserves-mass < `0.03`

runs: `default`

F-02 · contradicts · Plausible

Field-dependent creation in the co-simulation is exact: created A is 0.994× the exact expectation k2·[B]·V·602.214·t (creation-only, static B), confirming that using each node's actual cell volume (1/2–1/8 at boundaries) removes the overproduction; the 0.6% shortfall is the ~2-step Smoldyn creation start-up artifact.

from test: cosim-creation-is-exact `range`

runs: `default`

F-03 · contradicts · Plausible

The native VCell embedded solver diverges strongly and the divergence is fully explained by a known defect: native creation is 1.813× the expectation, matching the predicted full-volume boundary overproduction Σ(full)/Σ(actual)=1.815× to 0.1% (ratio 0.999). As a consequence native (A+B) grows 177% in 5 s and its steady-state ratio is 51% off, so the two solvers disagree at |z| ≈ 219 (totals) / 153 (profile). This is vcell-fvsolver's bug (virtualcell/vcell-fvsolver#24), not the coupling method.
from test: native-overproduction-matches-boundary-volume-defect `range` runs: `default`

Verdict

● PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-03): - **Co-simulation:** conserves A + B to 1.2%; steady-state ratio within 0.3% of k2/k1; creation at 0.994 of the exact expectation. - **Native VCell:** creation at 1.813× the expectation, matching the predicted boundary-volume overproduction (1.815×) to 0.1%. As a result A + B grows 177% in 5 s and the steady state is 51% off; the two solvers disagree strongly (|z| ~ 200). This is a defect in the embedded solver (virtualcell/vcell-fvsolver **24), not in the** coupling method. With field-dependent creation the co-simulation is the more accurate of the two.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.two_way_exchange` — 0 processes wired to 4 compartments. ● writes · ● reads.
`viva_pde_particle.composites.examples.two_way_exchange`

Simulation run

`default`

`cosim-conserves-mass` authored: PASS

`cosim-reaches-steady-state` authored: PASS

`cosim-creation-is-exact` authored: PASS

`native-overproduction-matches-boundary-volume-defect` authored: PASS

Reproduce: `vwb run study two-way-exchange`
cosim-two-way: `vwb run study two-way-exchange --variant cosim-two-way`
native-two-way: `vwb run study two-way-exchange --variant native-two-way`
creation-only: `vwb run study two-way-exchange --variant creation-only`

Tests

Max over time of |ensemble-mean (A + B) / initial − 1| for the co-simulation. — `< 0.03` pending

`cosim-conserves-mass · primary · derived_scalar`

Conserved in expectation; the residual comes from the explicit-coupling lag plus the ~2-step creation start-up artifact.

Relative error of the late-time (t ≥ 4 s) ratio of total A to total B vs k2/k1 for the co-simulation. — `< 0.03` pending

`cosim-reaches-steady-state · primary · derived_scalar`

Detailed balance of A ⇌ B; 16 seeds.

Created A / exact expectation for the co-simulation (creation only, static B). — `range undefined` pending

`cosim-creation-is-exact · primary · derived_scalar`

Exact expectation k2·[B]·V·602.214·t; the lower slack covers the ~2-step start-up artifact.

Native created-A ratio divided by Σ(full cells)/Σ(actual cells), the predicted overproduction of vcell-fvsolver's full-volume boundary creation. — `range undefined` pending

`native-overproduction-matches-boundary-volume-defect · secondary · derived_scalar`

Characterizes the reported defect (virtualcell/vcell-fvsolver#24). This test will fail once the defect is fixed upstream; then replace it with a mass-balance test for the native solver.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

04 A_particle + B_field → C_field: two-way nonlinear coupling, both solvers

ACCEPTED

When particles consume a field (B substantially depleted by A), do the co-simulation and the native VCell hybrid agree with each other and with the deterministic continuum limit?

Hypothesis & discriminating control

The particle side removes A at rate k·[B](x), and the PDE side removes k·[A_binned]·[B] from B and adds it to C. With 20,000 A, both solvers' ensemble means approach the continuum solution (A treated as a field). The solvers differ only by sampling. B + C is conserved exactly (PDE side).

MODEL EXPERIMENT 11×11×3 slab; 20,000 A (D = 1, Poisson initial placement in both solvers); B_0 = 0.5 µM (≈30,000 molecules, D = 1); C_0 = 0 (D = 1); k = 1/(µM·s); dt = 0.01; t = 3 s; 16 seeds per solver. Compared: - surviving fraction, produced C and final x-profiles (co-sim vs native); - survival vs the continuum FV solution.

OBSERVABLE `a4_cosim_vs_native_survival_max_z, a4_cosim_vs_native_c_max_z, a4_cosim_vs_native_profile_max_z, a4_cosim_vs_continuum_survival_max_abs_diff, a4_native_vs_continuum_survival_max_abs_diff…`

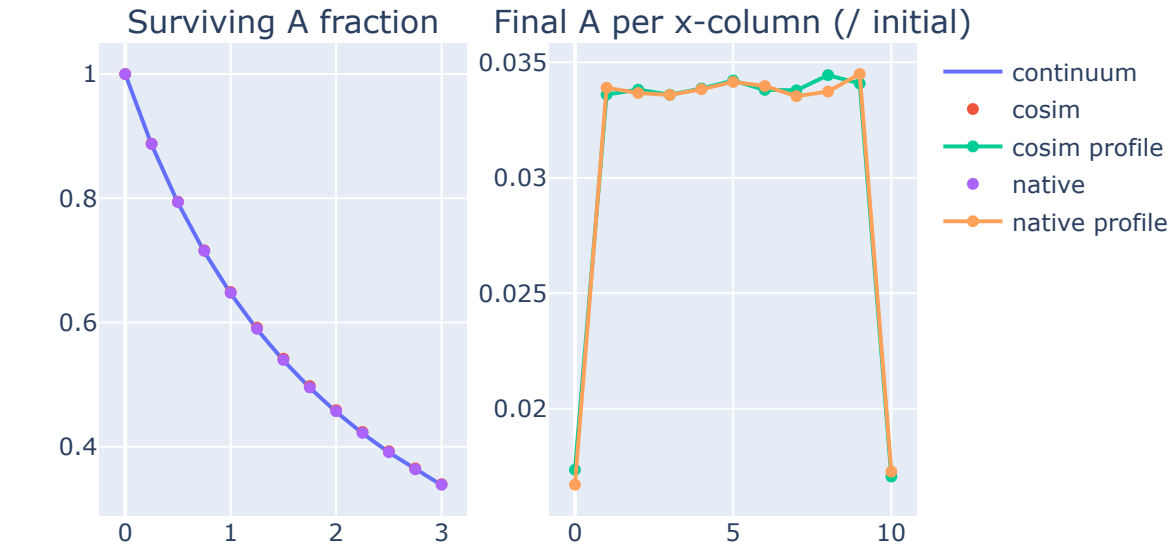
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓✓ 4/4 passed

Visualizations

BIMOLECULAR HYBRID



Finding

F-01 · confirms ● Observed

The co-simulation and the native VCell hybrid agree to within sampling on the two-way nonlinear coupling: max-over-time |z| is 1.79 for surviving A fraction, 0.85 for produced C, and 2.10 for the final x-profile — all well under the |z| < 4 threshold.

from test: solvers-agree-on-survival < 4

runs: `default`

F-02 · confirms ● Observed

Both solvers approach the deterministic continuum limit at 20,000 particles: the max-over-time absolute survival difference is 0.29% (co-sim) and 0.18% (native), both under the 1% band. The remainder is explicit-coupling lag plus discreteness.

from test: cosim-near-continuum < 0.01

runs: `default`

F-03 · confirms ● Observed

B + C is conserved to round-off on the PDE side in both solvers (max relative deviation 3.6e-14 co-sim, 3.2e-14 native), confirming the particle-consumption / field-production handoff is mass-balanced. Reaching this agreement exposed and fixed a float-sum record-time bug in process-bigraph (pre-fix survival |z| = 5.9, C |z| = 71).

from test: solvers-agree-on-product < 4

runs: `default`

Verdict

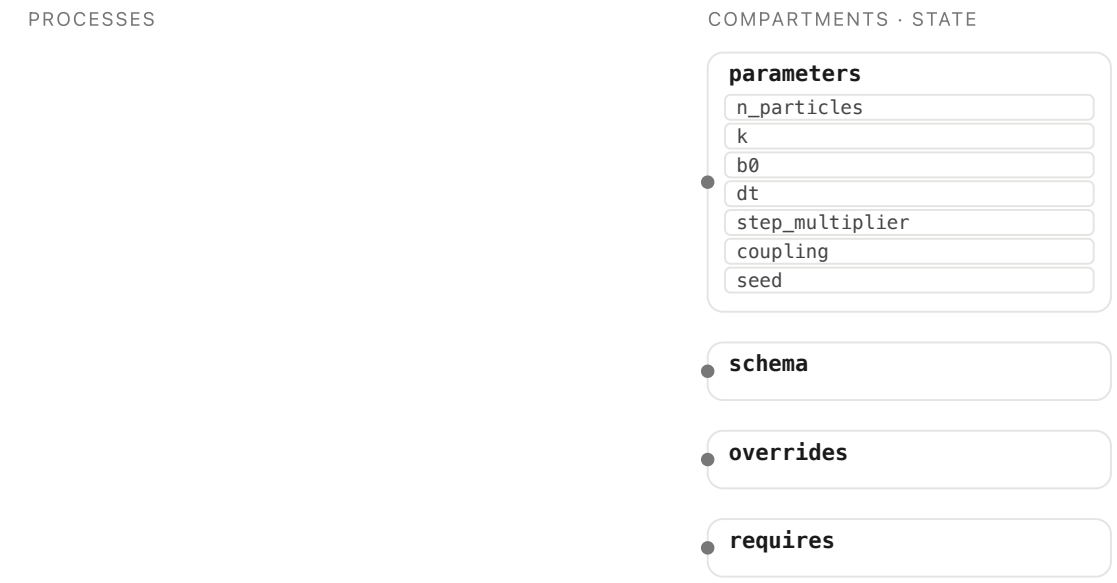
PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-03): - co-sim vs native: survival |z| = 1.79, produced C |z| = 0.85, final profiles |z| = 2.10; - vs the continuum: within 0.29% (co-sim) and 0.18% (native) absolute survival; - B + C conserved to round-off in both. This study exposed a harness bug, now fixed. Process-bigraph accumulates process times as float sums (25 × 0.01 > 0.25), so recorded co-sim states were one step behind the record time. Before the fix the solvers appeared to disagree on C (|z| = 71) and on survival (|z| = 5.9). run_hybrid now advances half a step past each record time, and all studies were re-run.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`solvers-agree-on-survival` authored: PASS

`solvers-agree-on-product` authored: PASS

`cosim-near-continuum` authored: PASS

`native-near-continuum` authored: PASS

Reproduce: `vwb run study bimolecular-hybrid`
native-vcell-arm: `vwb run study bimolecular-hybrid --variant native-vcell-arm`
continuum-limit: `vwb run study bimolecular-hybrid --variant continuum-limit`

Tests

Max over time of the two-sample |z| of the surviving A fraction, co-simulation vs native. `solvers-agree-on-survival · primary · derived_scalar` Same model, mesh and coupling; independent ensembles of 16.	< 4	pending
Max over time of the two-sample |z| of produced C (/ initial A), co-simulation vs native. `solvers-agree-on-product · primary · derived_scalar` C is produced on the PDE side from binned A; any timing or units mismatch shows up here first.	< 4	pending
Max over time of |survival fraction – continuum survival| for the co-simulation. `cosim-near-continuum · primary · derived_scalar` 20,000 molecules per run; the remainder is explicit-coupling lag plus discreteness.	< 0.01	pending
Max over time of |survival fraction – continuum survival| for the native VCell hybrid. `native-near-continuum · secondary · derived_scalar` As above.	< 0.01	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 05 Coupling error vs Smoldyn step multiplier, co-simulation (two schemes) and native VCell

● ACCEPTED

How does the error of the lagged explicit coupling grow with the coupling interval k·Δt (Smoldyn steps once per k PDE steps)? Does the co-simulation reproduce fvsolver's coupling error, and does process-bigraph's plain scheduling ("start-of-interval") do better or worse?

Hypothesis & discriminating control

Explicit lagged coupling is first order in k·Δt. The co-sim "fvsolver" mode applies the same lag as the embedded solver, so it has the same error. "start-of-interval" reads the field at the start of the Smoldyn step instead of at T+(k–1)Δt, so its error differs; its sign and size are to be measured.

MODEL EXPERIMENT A4 model with k = 2/(μM·s), 20,000 A, Δt = 0.005, t = 1.6 s, 64 seeds per configuration, step multiplier k ∈ {1, 2, 4, 8, 16, 32}. The error is the max over time of |mean surviving fraction – continuum limit| (A continuous, Δt/10). The order is fitted for k ≥ 4, above the sampling floor (~5·10⁻⁴).

OBSERVABLE `a5_cosim_fvsolver_order, a5_native_order, a5_fvsolver_mode_vs_native_err_ratio_k16, a5_cosim_fvsolver_err_k{1..32}, a5_cosim_start_of_interval_err_k{1..32} / a5_cosim_start_of_interval_order…`

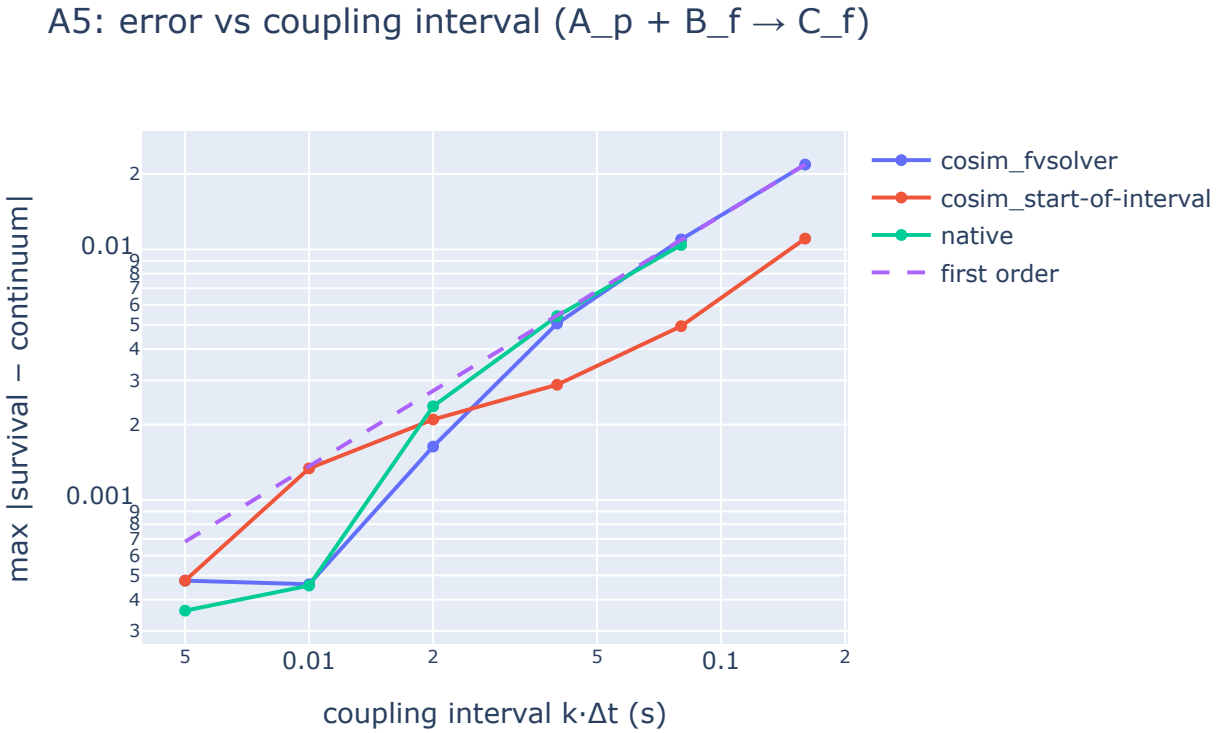
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓ 3/3 passed

Visualizations

COUPLING CONVERGENCE



Finding

F-01 · confirms ● Observed

The lagged explicit coupling is first order in the coupling interval k·Δt on both fvsolver-semantics solvers: fitted order (k ≥ 4) is 1.23 (co-sim fvsolver mode) and 1.07 (native), both inside the theoretical [0.7, 1.5] band.

from test: cosim-fvsolver-mode-first-order `range`

runs: `default`

F-02 · confirms · Observed

The co-simulation reproduces the embedded solver's coupling error, not just its results: at k = 16 (0.08 s) the co-sim fvsolver-mode error is 0.0110 and the native error 0.0104, a ratio of 1.05 (band [0.8, 1.25]).

from test: cosim-reproduces-native-coupling-error `range`

runs: `default`

F-03 · confirms · Observed

Plain process-bigraph "start-of-interval" scheduling roughly halves the coupling error at large k (0.0049 at k=16, 0.0110 at k=32 vs 0.0110 and 0.0218 for fvsolver mode; fitted order 0.80), because reading the field at the start of the particle step partly cancels the PDE side's lag. vcell-fvsolver's fixed ordering cannot use this without changing the solver.

from test: cosim-fvsolver-mode-first-order `range`

runs: `default`

Verdict

PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-03): - **Both fvsolver-semantics solvers are first order** in the coupling interval: fitted order 1.23 (co-sim fvsolver mode) and 1.07 (native). - **Their errors match:** at k = 16 (0.08 s) the co-sim error is 0.0110 and native 0.0104 (ratio 1.05). The co-simulation reproduces the embedded solver's coupling error, not only its results. - **"start-of-interval" coupling** (plain process-bigraph scheduling) roughly halves the error at large k: 0.0049 at k = 16 and 0.0110 at k = 32, against 0.0110 and 0.0218 for fvsolver mode. Reading the field at the start of the particle step appears to partly cancel the PDE side's lag. vcell-fvsolver's hybrid loop fixes one ordering, so that choice is not available there without changing the solver; higher-order splitting is Investigation B. - **Native crash:** VCell's native solver crashed at k = 32 (Smoldyn step 0.16 s, equal to the output interval), so native results stop at k = 16.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`cosim-fvsolver-mode-first-order` authored: PASS

`native-first-order` authored: PASS

`cosim-reproduces-native-coupling-error` authored: PASS

Reproduce: `vwb run study coupling-interval-convergence`
step-multiplier-sweep: `vwb run study coupling-interval-convergence --variant step-multiplier-sweep`
native-vcell-arm: `vwb run study coupling-interval-convergence --variant native-vcell-arm`

Tests

Fitted order (k ≥ 4) of the co-sim error with coupling="fvsolver". — range undefined — pending
`cosim-fvsolver-mode-first-order · primary · derived_scalar`
Explicit lagged operator splitting is first order in the coupling interval.

Fitted order (k ≥ 4) of the native VCell hybrid error. — range undefined — pending
`native-first-order · primary · derived_scalar`
Same scheme (SimTool.cpp main loop).

Co-sim (fvsolver mode) error / native error at k = 16. — range undefined — pending
`cosim-reproduces-native-coupling-error · primary · derived_scalar`
Same lag semantics (verified in tests/test_scheduling_semantics.py), so the same coupling error.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

06 Wall time of the co-simulation vs the native VCell hybrid, with breakdown

ACCEPTED

What does decomposing the hybrid solver into process-bigraph processes cost? How does the co-simulation's wall time compare with the native embedded solver as the grid and the particle count grow, and how much of it is framework overhead?

Hypothesis & discriminating control

The co-simulation pays a per-step framework overhead (scheduling, state updates and data marshaling between processes) that matters most for small problems. For large problems, PDE and particle work dominate in both implementations.

MODEL EXPERIMENT A4 model (A_p + B_f → C_f) on N×N×3 slabs, dt = 0.01, t = 1 s (100 steps), one process, median of 3 runs. Sweeps: - N ∈ {11, 21, 41, 81} with 20,000 particles; - particles ∈ {$2{\cdot}10^3$, $2{\cdot}10^4$, $2{\cdot}10^5$} on 21×21×3. Co-sim time is split into PDE update, particle update and framework (the remainder). Native time is split into input generation, solve and read.

OBSERVABLE `a6_cosim_framework_ms_per_step_max`, `a6_grid_n81_cosim_over_native_solve`, `a6_grid_n{11,21,41,81}_cosim_s`, `a6_grid_n{11,21,41,81}_cosim_{framework,particle,pde}_frac`, `a6_particles_p{2000,20000,200000}_cosim_s…`

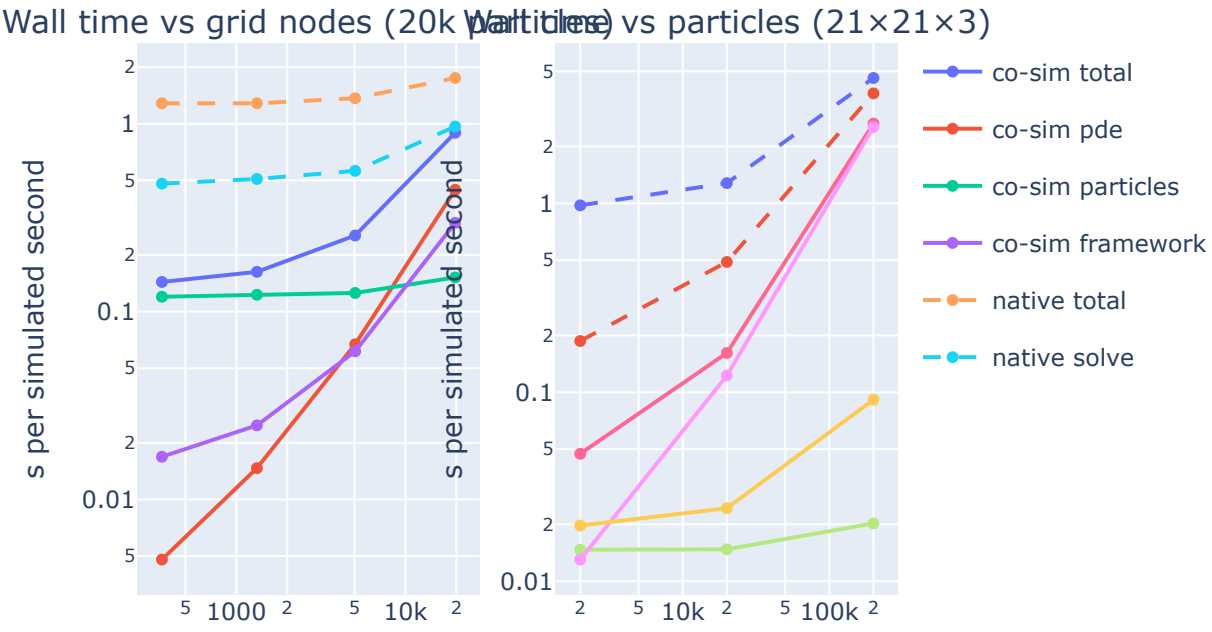
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓ 2/2 passed

Visualizations

PERFORMANCE



Finding

F-01 · confirms ● Observed

The process-bigraph framework overhead is bounded and small: at most 2.97 milliseconds per step across all configurations (the bound is 10 milliseconds), ranging from 3.5% of co-sim time on the heaviest problem (200k particles) up to 41.7% on the smallest (2k particles on 21×21×3), where there is little real work to amortize it against.

from test: framework-overhead-bounded < `10`

runs: `default`

F-02 · confirms ● Observed

The modular co-simulation is faster than the native embedded solve in every configuration: co-sim/native-solve ranges from 0.25× (2k particles) to 0.93× on the largest grid (81×81×3), and 0.69× at 200k particles (co-sim 2.64 s vs native solve 3.82 s). The decomposition costs nothing at scale — well under the 2× bound.

from test: cosim-not-slower-than-native-at-scale < `2`

runs: `default`

F-03 · confirms ● Observed

Co-sim time shifts cleanly with the workload: particles dominate at high particle counts (95.9% of co-sim time at 200k particles, PDE only 0.8%), while the PDE dominates on the largest grid (49.7% at 81×81×3). Framework overhead is largest

precisely where total work is smallest (41.7% at 2k particles), as expected for a fixed per-step scheduling cost.

from test: framework-overhead-bounded < `10`

runs: `default`

Verdict

PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-03; Apple Silicon, single process): - **The co-simulation is faster than the native solve** in every configuration: 0.25–0.93× of the native solve time alone. Native total time also includes about 0.5 s of input generation (libvcell math generation) and a process spawn. - Grid sweep, 20k particles: co-sim 0.14 s → 0.90 s; native solve 0.48 s → 0.97 s. - 200k particles: co-sim 2.6 s, native solve 3.8 s. - Likely causes: modern Smoldyn 2.7x vs 2.38, the direct sparse LU in FVReactionDiffusion vs VCell's PCG, and VCell's per-output file writes. - **Framework overhead** is at most 3.0 ms per step: 3–42% of co-sim time, largest for the smallest problems (2k particles on 21×21×3). - **Where the co-sim time goes:** particles dominate at high particle counts (96%); the PDE dominates on the largest grid (50%). The modular co-simulation costs nothing in performance on these problems, and is in fact faster. The overhead to watch is process-bigraph's per-step scheduling, for very small, many-step problems.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

framework-overhead-bounded authored: PASS

cosim-not-slower-than-native-at-scale authored: PASS

Reproduce: `vwb run study performance-scaling`

grid-sweep: `vwb run study performance-scaling --variant grid-sweep`

particle-sweep: `vwb run study performance-scaling --variant particle-sweep`

Tests

Largest co-sim framework overhead per step (ms) across all configurations. `< 10` pending

`framework-overhead-bounded · primary · derived_scalar`

A two-process composite should cost no more than a few milliseconds per step to schedule and marshal.

Co-sim total time / native solve time on the largest grid (81×81×3). `< 2` pending

`cosim-not-slower-than-native-at-scale · secondary · derived_scalar`

The modular decomposition should not cost more than 2× the embedded solver at scale.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

Synthesis & gaps

Where results agree, conflict, or leave open claims — and the priority follow-ups that would resolve them.

Contradictions & uncertainty

FINDING CONFLICT **— Mass balance across the particle ↔ field handoff**

Field-dependent creation in the co-simulation is exact: created A is 0.994× the exact expectation k2·[B]·V·602.214·t (creation-only, static B), confirming that using each node's actual cell volume (1/2–1/8 at boundaries) removes the overproduction; the 0.6% shortfall is the ~2-step Smoldyn creation

Mass balance across the particle ↔ field handoff SUPPORTED

FINDING CONFLICT **— Mass balance across the particle ↔ field handoff**

The native VCell embedded solver diverges strongly and the divergence is fully explained by a known defect: native creation is 1.813× the expectation, matching the predicted full-volume boundary overproduction Σ(full)/Σ(actual)=1.815× to 0.1% (ratio 0.999).

Mass balance across the particle ↔ field handoff SUPPORTED

CAUSAL CONTINUITY

These studies take their inputs from outside the chain (a supplied feed or reset state) rather than from an upstream study's output — so their result does not yet rest on a modeled upstream mechanism:

Each side alone, and the particle-to-grid handoff

Validation & reproducibility

Automated, code-computed consistency checks (no AI): conservation/invariant ledgers, computed-vs-authored verdicts, run reproducibility (environment + determinism), parameter drift and acceptance-criterion coverage. Each study's own ledger predicates (e.g. mass/charge balance) are its gating behavior tests, listed on the study's card (its Tests).

> **Stale finding** `F-01`
> Finding 'F-01' declared a next_action but no study was seeded

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> Finding 'F-02' declared a next_action but no study was seeded

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> **Stale finding** `F-03`
> Finding 'F-03' declared a next_action but no study was seeded

> **Stale finding** `F-03`
> Finding 'F-03' declared a next_action but no study was seeded

> **Stale finding** `F-03`

Finding 'F-03' declared a next_action but no study was seeded

Stale finding `F-03`

Finding 'F-03' declared a next_action but no study was seeded

Stale finding `F-03`

Finding 'F-03' declared a next_action but no study was seeded

▼ REPORT READINESS — 122 FINDINGS ACROSS 11 CHECKS (0 ERRORS, 122 WARNINGS)

44 band test missing cites warning

`cosim-boundary-transfer, dumbbell-three-ways, fenicsx-same-problem, handoff-units-and-binning +8`

18 runs without emitter warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +14`

18 unresolved composite warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +14`

14 dag edges legacy only warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +10`

6 authored confidence diverges warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, single-channel-fokker-planck +2`

6 investigation narrative spine required warning

`cosim-vs-embedded-hybrid, hybrid-generalizability, vcell-hybrid-paper-benchmarks`

5 baseline config carry warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, single-channel-fokker-planck +1`

5 reviewer clarity ambiguity warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, sphere-cosim-vs-native +1`

3 investigation missing adversarial warning

`cosim-vs-embedded-hybrid, hybrid-generalizability, vcell-hybrid-paper-benchmarks`

2 unearned unit labels warning

`near-membrane-fields, sphere-cosim-vs-native`

1 dag edges legacy redundant warning

`field-modulated-decay`

Evidence register

Every study in the campaign (6) — scientific verdict, run status, and one-line result, each linked to its full study card above.

STUDY	Verdict	Run	Result
Each side alone, and the particle-to-grid handoff	SUPPORTED	ran	Each half of the co-simulation is correct in isolation: the FV operator reproduces the analytic cosine-mode decay to 6.0e-4 relative error, and Smoldyn's binned per-axis variance (dx²/6-corrected) matches 2Dt to 0.13% re
Field-to-particle coupling, A_particle -> 0 at rate k·[B](x)	SUPPORTED	ran	The particle side reads the PDE field correctly: with static B and immobile A, per- column survival matches N0·exp(-k·B_col·t) with max |z| = 1.56 (co-sim) and 1.68 (native VCell), both well under the |z| < 4 band, and e
Mass balance across the particle ↔ field handoff	SUPPORTED	ran	The co-simulation conserves mass and reaches the right steady state across the two-way handoff: ensemble-mean (A+B) stays within 1.18% of its initial value over 5 s, and the late-time total-A/total-B ratio is within 0.32
A_particle + B_field → C_field: two-way nonlinear coupling, both solvers	SUPPORTED	ran	The co-simulation and the native VCell hybrid agree to within sampling on the two-way nonlinear coupling: max-over-time |z| is 1.79 for surviving A fraction, 0.85 for produced C, and 2.10 for the final x-profile — all we
Coupling error vs Smoldyn step multiplier, co-simulation (two schemes) and native VCell	SUPPORTED	ran	The lagged explicit coupling is first order in the coupling interval k·Δt on both fvsolver-semantics solvers: fitted order (k ≥ 4) is 1.23 (co-sim fvsolver mode) and 1.07 (native), both inside the theoretical [0.7, 1.5]
Wall time of the co-simulation vs the native VCell hybrid, with breakdown	SUPPORTED	ran	The process-bigraph framework overhead is bounded and small: at most 2.97 milliseconds per step across all configurations (the bound is 10 milliseconds), ranging from 3.5% of co-sim time on the heaviest problem (200k par

Leave an overall note, and per-study notes on each study card. Then **Review & export** assembles them into a YAML feedback report.

FEEDBACK · THE WHOLE REPORT 👍 Looks right ✏️ Needs revision

Add a note...

Rendered from workspace `viva-pde-particle--outcomes` · deterministic from investigation + study YAML.

`investigations/cosim-vs-embedded-hybrid/investigation.yaml · studies/*/study.yaml · loop-trajectory JSON`